

Informational Energetics: Universal Architecture and the Entropic Action of Persistence

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All persistent systems must solve a universal problem: maintaining structural coherence against entropic erasure. We present Informational Energetics, a substrate-independent framework deriving the minimal architecture of persistence at zero flux from information-theoretic first principles. We demonstrate that any persistent system must solve three existential problems: Finiteness (bounded resources), Conservation (preserved information), and Causality (well-defined evolution), yielding six irreducible pillars: Capacity, Map, Protocol, Governor, Toll, and Margin. Necessity is established by counterfactual elimination, and sufficiency by compositional reduction. We ground this static architecture in time by deriving the minimum dynamical description from three irreducible primitives of state transition: irreducible toll (ξ), structural complexity density ($\mathbf{K}$), and coordinated transition flux (f). Their product is the dissipation density $\mathcal{D} = \xi \cdot \mathbf{K} \cdot f$. We derive the Entropic Action S_Φ , the cumulative spacetime integral of $\mathcal{D}$, and demonstrate that the Principle of Least Action emerges as the boundary of survivorship: non-minimizing trajectories undergo exponential suppression, leaving only the least-action remainder as observable history. Mapping diverse systems onto this architectural invariant recasts domain-specific heuristics as instances of the same universal mechanics: solutions validated in one domain transfer to another, persistent systems become directly comparable, and structural failure becomes a quantifiable, cross-domain phenomenon.

I. INTRODUCTION

From biological organisms to engineered feedback loops, complex systems science describes how structures maintain coherence against entropic disturbances. These structures share a defining property: they *persist* against the current that erases distinction. Control theory describes how systems regulate against fluctuations, taking the regulation target (setpoint) as an externally specified boundary condition. Information theory quantifies the cost of information processing, taking the processor as given. Thermodynamics establishes the arrow of time, treating entropy as a property of states. These three fields developed independently to solve distinct engineering problems: thermodynamics from heat engines, information theory from telecommunication, and control theory from servomechanisms. Each optimizes systems already embedded in a supportive environment.

Here, we address the preceding gap: the minimal architecture required before optimization or dynamics can even be written. Grounding this architecture in algorithmic information theory, IE functions as an abstraction layer beneath thermodynamics and control theory, which emerge as substrate-specific instantiations of the same invariant.

Previous complex systems research has converged on pieces of this puzzle from multiple directions. Prigogine established that persistent structures occupy a minimum-entropy-production attractor[1]; England showed how external drive selects for structural configurations in non-equilibrium steady states[2]; and cybernetics has cata-

logued the organizational patterns, such as requisite variety and good regulation, that correlate with survival[3, 4]. Individual domains have independently named the specific components that perform these same functions, without recognizing them as instances of one shared architecture. What these frameworks share is that they treat the persistent system, or its named components, as given: they describe how a structure maintains itself, adapts, or is selected, but do not formally define the minimal architecture required before any of those dynamics can begin.

IE consists of two universal components. First, the **Architecture of Persistence** (section II): six substrate-independent pillars extracted from the logical requirements of Finiteness, Conservation, and Causality. This architecture specifies what must exist, not how it behaves under transition. Second, the **Dynamics of Persistence**(section III): the universal behavior governing how any persistent system dissipates, accumulates entropic action, and undergoes selection. We derive the dissipation density $\mathcal{D} = \xi \cdot \mathbf{K} \cdot f$ as the minimum dynamical description any state transition law must presuppose.

Section IV demonstrates that the Principle of Least Action emerges as the boundary of survivorship: trajectories that do not minimize their Entropic Action undergo exponential suppression, leaving only the least-action remainder as observable history. Section VI then illustrates the architecture’s mapping across biological, ecological, and computational substrates.

II. THE ARCHITECTURE OF PERSISTENCE

Any system that maintains structural coherence against entropic decay must solve three existential prob-

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lems. The modes of engagement with these problems yield six irreducible pillars, which we derive, quantify, and prove minimally necessary and sufficient by logical exhaustive partitioning.

A. The Three Existential Problems

Any process persisting through time is bounded in extent, carries distinguishable content, and evolves through an ordered sequence of states. Absent any one, it is indistinguishable from its environment, from noise, or from something static. **Finiteness** (bounded resources against divergence), **Conservation** (preserved information against dissipation), and **Causality** (well-defined temporal evolution against ambiguity) name the corresponding failure mode of each, exhausting what it means for something to be a bounded, stateful, evolving process. Each of these demands two modes of structural engagement: **provision** (the resource must exist as an allocated stock) and **defense** (the allocation must survive its characteristic perturbation).

	Provision	Defense
Finiteness	Capacity	Margin
Conservation	Map	Protocol
Causality	Governor	Toll

Table I. The three existential problems and their two modes of structural engagement yield the six irreducible pillars of persistence.

These two modes are exhaustive: a resource either exists or it does not, and if it exists, it either survives its characteristic perturbation or it does not. A resource that fails to exist cannot be engaged at all, and one that exists but does not survive is indistinguishable from one that never existed; provision and defense are therefore the only two structural relations a system can hold to a requirement, leaving no third mode.

B. The Six Pillars

Each pillar is independently attested by a convergent theorem from a distinct discipline, subordinating performance to persistence.

1. **Capacity (CAP): The Vessel.** The infrastructure required to acquire resources, process signals, and perform work. It defines the system’s operational potential. *Convergent Evidence (The Bekenstein Bound [5]):* Bekenstein’s bound limits the information content of any finite region, and the Margolus-Levitin limit[6] bounds the rate of state transitions; together they establish that finite capacity is a structural invariant.

2. **Map (MAP): The Model.** The internal logic, topological structure, or compressed representation of environmental regularities that distinguishes the system from the environment. It functions as a predictive engine that reduces uncertainty (informational surprise), allowing the system to steer towards coherent states. *Convergent Evidence (The Good-Regulator Theorem [4]):* Conant and Ashby’s Good-Regulator theorem establishes that persistent regulation requires an internal model; without Map, the system–environment boundary dissolves.
3. **Protocol (PRO): The Interface.** The communicative glue regulating information flow between the Capacity and the Map. It ensures internal coherence, standardizes interfaces, and minimizes signal decay across the system’s internal boundaries. *Convergent Evidence (Shannon’s Source-Channel Separation and Channel Capacity [7]):* Shannon’s source–channel separation theorem shows that coherence across internal boundaries requires an independent interface layer; Protocol is that layer.
4. **Governor (GOV): The Constraint.** The non-linear constraint mechanism that prevents unbounded divergence or ‘runaway’ loops. It enforces operational boundaries (e.g., apoptosis, budget limits) to protect the substrate from exhaustion. *Convergent Evidence (Ashby’s Law of Requisite Variety [8]):* Ashby’s Law of Requisite Variety requires that the constraint mechanism match the variety of perturbations; without Governor, runaway is unavoidable.
5. **Toll (TOL): The Temporal Cost.** State transitions carry an irreducible information cost (Landauer’s principle) and an irreducible temporal cost (Margolus-Levitin limit). Any irreversible logical operation, such as erasing a bit of structural information, necessitates a strictly compensatory increase in environmental Shannon entropy (the $k_B T \ln 2$ bound). *Convergent Evidence (Landauer’s Principle and Margolus-Levitin Limit [6, 9]):* These bounds make Toll a structural requirement.
6. **Margin (MAR): The Buffer.** The structural and energetic reserves held to absorb stochastic shocks. It defines the system’s ‘Safe Operating Area,’ providing the buffer required to survive environmental variance without terminal failure. (The gain and phase margin). *Convergent Evidence (Nyquist-Bode Stability Margins[10] and Kingman’s Formula[11]):* A system operating at theoretical criticality has zero tolerance for variance. Kingman’s formula shows that as a queueing system’s utilization approaches unity, waiting time diverges asymptotically, zero buffer against arrival variance means unbounded delay. Nyquist’s stability margins are the linear-control manifestation of the

same general requirement: persistent systems must maintain structural tolerance away from critical thresholds.

Every system operates upon a **Substrate**: the medium imposing immutable limits (bandwidth, latency, noise floor) within which the six pillars function. IE concerns closed-loop systems that maintain coherence against entropic pressure. This is not agency in any teleological sense; it is a **structural filter**. Only configurations implementing the six-pillar architecture survive observation; all others dissipate.

IE describes the architecture of persistence but does not explain the transition from disorder to structure.

C. The Entropic Balance Sheet

A persistent system operates as a dissipative resistor against entropic erosion. Its total entropic impedance Z_{IE} is the sum of all structural costs required to maintain the six-pillar architecture. Every pillar imposes a strictly positive cost: a system cannot maintain boundaries, store internal models, or execute state updates for free. Within this strictly positive budget, the pillars partition conceptually into two distinct operational roles:

- **Direct Structural Costs:** The irreducible physical overhead of maintaining the system's baseline existence. Maintaining the vessel (Z_{CAP} , Z_{MAR}), the internal model (Z_{MAP}), and executing state updates (Z_{TOL}) requires a strict allocation of resources regardless of how the system is organized.
- **Coordination Costs:** Protocol and Governor incur direct structural costs to operate, but they function as active constraints rather than passive structures. Their operational effect is to suppress uncoordinated variance.

The system pays a continuous *coordination cost* (maintaining Protocol and Governor) as an insurance premium to avoid a catastrophic entropic penalty (fragmentation or runaway divergence). In control theory, this is analogous to negative feedback: the active controller consumes continuous energy to operate, but its regulation ensures the closed loop's total dissipation is lower than the open-loop alternative.

In Quiescent Steady State, the six pillars operate on functionally orthogonal domains. The Net Entropic Impedance is therefore a strictly additive budget across these independent channels:

$$Z_{IE} = Z_{CAP} + Z_{MAP} + Z_{TOL} + Z_{MAR} + Z_{PRO} + Z_{GOV} \quad (1)$$

Because the pillars occupy orthogonal domains at zero flux, any correlation between them is already priced into the individual coordination terms Z_{PRO} and Z_{GOV} ;

no additional cross-term is needed to capture coupling between pillars within this baseline. Coupling outside Quiescent Steady State introduces non-linear contention, which we formalize in section V.

D. Necessity: Counterfactual Failure Modes

The necessity of the set $\mathbb{P} = \{CAP, MAP, PRO, GOV, TOL, MAR\}$ follows by analyzing the **Counterfactual Failure Mode** of a system lacking exactly one pillar ($\mathbb{P}' = \mathbb{P} \setminus \{x\}$). In every case, the expected lifetime of the system τ tends to zero.

- **No Capacity (Starvation):** Lacks an energetic envelope to respond to perturbations; system lifetime drops $\tau \rightarrow 0$ via rapid exhaustion.
- **No Map (Dissolution):** Lacks self-definition; random activation maximizes internal uncertainty. $\tau \rightarrow 0$ as the system-environment boundary dissolves.
- **No Protocol (Decoherence):** Components decouple; actions rely on statistically independent, outdated states. $\tau \rightarrow 0$ by organizational fragmentation.
- **No Governor (Divergence):** Unconstrained positive feedback exceeds structural limits. $\tau \rightarrow 0$ by unbounded runaway.
- **No Toll (Zeno Collapse):** State transitions require irreducible time and energy per the Margolus-Levitin limit[6]. Without this cost, updates occur instantaneously, collapsing the causal sequence. $\tau \rightarrow 0$ by causal collapse.
- **No Margin (Fragility):** Operating at theoretical criticality leaves zero buffer for environmental variance. $\tau \rightarrow 0$ by random shock.

Since the removal of any component results in termination, every element of $\mathbb{P}$ is necessary.

E. Sufficiency: System Preservation

A seventh requirement could only arise from an external source or from internal composition. At zero flux, by definition, no external source exists. Composition is closed by the same taxonomy that generated $\mathbb{P}$: Section II A established that every persistence-relevant demand is one of three existential problems engaged through one of two modes, provision or defense. A composite capability cannot introduce a third mode or a fourth problem; it can only combine pillars already drawn from the six pillars. With both routes closed, the six pillars exhaust the minimal architecture at the zero flux baseline.

Capabilities traditionally treated as independent requirements bear this out. Memory is Map content provisioned against temporal variance by Margin. Prediction is Map operation on incoming signal via Protocol. Repair is Governor-triggered re-provisioning within the Margin reserve. Replication is Protocol-mediated write of Map onto fresh Capacity. Advanced topologies, such as feedforward networks and state observers, extend the six-pillar structure the same way: none introduces a new primitive, each is a composition within the existing six.

This exhaustion extends to open systems, not merely the zero flux baseline. External energy or information flux raises the effective dissipation ceiling and modifies the balance sheet, but it does not eliminate the need for any pillar, nor open a route to a seventh: flux changes the budget, not the architecture that budget maintains. The zero flux baseline is the vacuum limit of persistence, the architecture stripped of environmental complication, not a special case exempt from the open-system result.

F. Relation to Existing Frameworks

The six pillars are not novel inventions; previous frameworks encountered pieces of the six-pillar architecture already. In each case the components were assumed to exist and not recognized as instances of one invariant beyond the framework's own domain. IE supplies what each was missing: a proof of why exactly these components are required, and the recognition that domains studying them in isolation have been studying the same architecture all along.

1. Relation to Information Theory

Channel optimization repeatedly encounters parameters corresponding to the six pillars: channel capacity (Capacity), codebook topology (Map), handshaking and framing (Protocol), power constraints (Governor), processing overhead (Toll), and link margins (Margin). Classical information theory treats these six as arbitrary external boundary conditions and optimizes throughput conditioned on them, without asking why a physical system must instantiate all six simply to persist against entropic erasure. IE exposes these engineering constraints as the irreducible, necessary architecture required for the channel hardware itself to exist, not tunables to optimize.

2. Relation to Thermodynamics

Thermodynamics assumes an external thermal bath and continuous ensemble variables from the outset; it defines temperature and equilibrium against that bath without asking what structure must exist prior to it. Algorithmic information theory supplies the missing formal substrate. Shannon entropy and Gibbs entropy

are formally identical in the statistical-mechanical limit, while Landauer's principle and the Margolus-Levitin bound supply the physical scale of individual state transitions: information loss corresponds to entropic dissipation, channel capacity to thermodynamic availability, and information flux to energy flux.

The difference is one of scope and substrate. IE requires neither bath nor temperature. It is defined at zero flux on a discrete information substrate, and thermodynamic phenomenology emerges as the continuum limit of accumulated discrete transitions. Where thermodynamics describes how heat flows, IE describes what structure must exist before heat or flow can be meaningfully defined.

3. Relation to Control Theory

Classical control theory emerged from engineering physical servomechanisms: radar mounts, autopilots, and chemical plants whose historical context was optimizing systems already built, powered, and staffed. Its four given components (the plant, the sensor, the controller, and the actuator) are convergent evidence for four of the six pillars: Plant/Actuator $\rightarrow$ Capacity, Reference Signal/Setpoint $\rightarrow$ Map, Sensor/Feedback Path $\rightarrow$ Protocol, Controller/Limiter $\rightarrow$ Governor.

The remaining two pillars are rarely canonized alongside the first four as they are not physical components, but describe how the four components behave under time and stress. Time delays, bandwidth limits, and processing latency ($\rightarrow$ Toll) were minimized as engineering overhead and gain/phase margins ($\rightarrow$ Margin) were tuned for stability.

Control theory asks how to regulate an existing system; IE asks what architecture makes existence possible. The classical four-component loop is not a rival account but one physical instantiation of the six-pillar invariant.

III. THE DYNAMICS OF PERSISTENCE

The static architecture of persistence necessitates a minimal dynamical description. A system that persists maintains structural coherence against entropic erosion; therefore it must export internal entropy to its environment at a rate that balances what it produces. We construct the measure of that dissipation from the three irreducible primitives that any dynamical law governing persistent systems must presuppose, prove their necessity, and integrate them into a cumulative Entropic Action that closes the loop back to the six-pillar architecture.

A. The Cost of State Transitions

While a system's architecture provides the capability to persist, the operation of those pillars determines how

long it actually persists. Any state transition does work, and all work generates entropy; left unconstrained, a system's own activity can result in its rapid dissolution.

Following Prigogine[1], persistence therefore requires two behaviors: internal entropy production must be exported to the environment fast enough that the system's own entropy never rises, and that production rate itself settles to the smallest value consistent with holding steady state ($\delta\sigma = 0$), below this floor a system is not yet stable, and above it, every transition is spending more than persistence strictly requires.

We rigorously quantify this at its limiting case, **Quiescent Steady State**: no net energy or information enters or leaves, Z_{IE} is constant, and the entropic balance sheet (section IIC) is strictly additive. The dynamic evolution of open systems is left to future work.

B. The Dissipation Density

Prigogine's Minimum Entropy Production theorem establishes *that* a persistent system must minimize its entropy production rate, but not *what* defines that rate. We derive the minimum dynamical description by identifying the irreducible primitives that any state transition law must presuppose.

When the static six-pillar architecture moves through time, any state transition (a local dissipative event) implicates three candidate functional dimensions: a subject to transition, an energetic toll to execute the transition, and a temporal frequency of execution. These are not assigned to individual pillars, but are properties of the transition itself.

1. **A structured payload (The Subject)**: Because a persistent system maintains a specific configuration rather than a statistical ensemble, its local state is quantified by its structural complexity $\mathbf{K}$. This is the dimensionless Kolmogorov complexity (as an uncomputable lower bound, for which computable proxies exist in applied use¹) of the local configuration relative to the substrate's reference encoding. *Convergent Evidence (Kolmogorov Complexity [12])*: The algorithmic information content of a state is the irreducible payload carried by any transition.

2. **An irreducible toll (The Cost)**: Every transition exacts a cost, measured as a dimensionless

count of elementary logical operations ξ . *Convergent Evidence (Landauer's Principle [9])*: Every irreversible logical operation carries a thermodynamic cost of at least $k_B T \ln 2$.

3. **A coordinated rate (The Frequency)**: State updates must occur at a transition flux f , a rate density of state updates per unit substrate measure. *Convergent Evidence (The Nyquist-Shannon Sampling Theorem [7])*: A signal can only be reconstructed without loss if sampled at a rate exceeding its bandwidth; below that rate, information is irrecoverably lost to aliasing. f is the substrate-level analogue of this minimum sampling rate, required to track the Map against noise.

The pillars operate on orthogonal domains (section IIC) in the Quiescent Steady State; $\mathbf{K}$, ξ , and f inherit this independence, each drawing on a distinct domain: algorithmic description, informational toll, and temporal rate, respectively. Because a local dissipation metric must yield a uniform density of elementary transitions per unit substrate measure, its functional form is dimensionally constrained. With $\mathbf{K}$ as [logical operations / local state], ξ as [dimensionless action units / logical operation], and f as [local states updated / (substrate spatial measure $\cdot$ time)]. Their strictly linear product yields $\mathcal{D}$ as [action units / (substrate spatial measure $\cdot$ time)].

The local **Dissipation Density** is therefore:

$$\mathcal{D} = \xi \cdot \mathbf{K} \cdot f \quad (2)$$

C. Necessity: Counterfactual Failure Modes

The necessity of $(\xi, \mathbf{K}, f)$ follows by the same counterfactual method used for the pillars (section IID): removing any factor drives lifetime to zero, as the local manifestation of its corresponding pillar failure mode.

- **No ξ (Zeno Collapse)**: Transitions become thermodynamically free, so the system attempts infinitely many in finite time, collapsing the causal sequence. $\tau \rightarrow 0$, the local form of No-Toll.
- **No $\mathbf{K}$ (Dissolution)**: No distinguishable local states remain; random activation maximizes uncertainty and the system-environment boundary dissolves. $\tau \rightarrow 0$, the local form of No-Map.
- **No f (Decoherence/Divergence)**: The system freezes into a static configuration, decoupled from its own history and unable to mount a defensive response. $\tau \rightarrow 0$, the local form of No-Protocol/No-Governor.

Since the removal of any factor results in termination, the set $(\xi, \mathbf{K}, f)$ is necessary.

¹ Uncomputability does not preclude operational use: the algorithmic-information-theory literature routinely substitutes computable proxies: Minimum Description Length, Lempel-Ziv complexity, normalized compression distance as finite-sample estimators of $\mathbf{K}$, a practice this paper's baseline derivation does not require but that applied instantiations of IE can draw on directly.

D. Necessity of the Product Form

The product form of equation (2) is the only functional form consistent with the counterfactual failure modes. An additive form would permit non-zero dissipation from any one factor alone (transition cost with no distinguishable states ($\mathbf{K} = 0$), or structure with no transitions ($f = 0$)) configurations already shown to have lifetime zero. Only the product form vanishes correctly when any factor is absent, matching the requirement that dissipation cannot occur without all three simultaneously.

E. Sufficiency: No Fourth Factor

Given the multiplicative form, a missing factor would drive $\mathcal{D}$ to zero rather than merely reduce it, leaving no independent dimension for a fourth factor to occupy. Any property proposed as a fourth factor, on inspection, either reduces to a combination of these three or describes something about the environment rather than the system itself.

F. The Complexity Trade-off

The product form of the Dissipation Density implies a finite-budget competition among the three factors.

Proposition 1 (The Complexity Trade-off). *For any system in Quiescent Steady State, the substrate's finite Capacity imposes a dissipation ceiling $\mathcal{D}_{\max}$:*

$$\xi \cdot \mathbf{K} \cdot f \leq \mathcal{D}_{\max} \quad (3)$$

No system can simultaneously maximize structural complexity, operational speed, and processing efficiency: increasing any one factor requires a proportional decrease in the product of the other two.

Proof. Exceeding $\mathcal{D}_{\max}$ at any point violates the stationary condition $\delta\sigma = 0$ (section III A); substituting the unique product form (equation (2)) into this ceiling yields the inequality directly. $\square$

A biological cell illustrates the binding constraint: ATP cost per operation (ξ), proteome complexity ($\mathbf{K}$), and metabolic turnover (f) together cannot exceed the thermodynamic ceiling of the cellular volume. A cell cannot maximize proteome complexity and reaction frequency at once without $\mathcal{D}$ diverging.

Convergent Evidence: As established for Capacity (section II B), the Margolus-Levitin limit and Bekenstein bound jointly cap the achievable volume in $(\xi, \mathbf{K}, f)$ -space, one bounding f for any finite ξ , the other bounding $\mathbf{K}$.

G. The Entropic Action

Having established the conditions for structural survival, we need a way to compare surviving systems by lifetime. A system's viability cannot be judged pointwise: no single pillar sustains a transition alone, so no single local event determines survival either, what accumulates across the whole system is what matters. We build that accumulated quantity in two steps: first collapsing the system's spatial extent into one number at each instant, then accumulating that number over its lifetime.

$\mathcal{D}$'s multiplicative, local form and Z_{IE} 's additive, global one (section II C) are reconciled by integrating over the system's spatial footprint $\Sigma(t)$: many local multiplicative events, summed, become one global additive quantity at each instant:

$$Z_{\text{IE}}(t) = \int_{\Sigma(t)} \mathcal{D} d\mu \quad (4)$$

with $d\mu$ the counting measure on the discrete substrate state space, a sum over discrete points, not a continuous integral.

With the impedance $Z_{\text{IE}}(t)$ known to be the system's total instantaneous dissipation, the cumulative **Entropic Action** is the time integral of that impedance over the system's lifetime:

$$S_{\Phi} = \int_0^{\tau} Z_{\text{IE}}(t) dt \quad (5)$$

1. Physical Meaning of the Entropic Action

Because $\mathcal{D}$ counts elementary state transitions per unit substrate measure, S_{Φ} is a pure, dimensionless count of irreversible transitions, not yet an energy. When IE is projected onto a thermal physical substrate, Landauer's principle supplies the thermodynamic scale $\xi_0 = k_B T \ln 2$, yielding the physical dissipation analogue $E_{\text{diss}} = \xi_0 S_{\Phi}$. On athermal substrates, S_{Φ} remains well-defined; only the conversion factor to macroscopic energy changes.

2. Necessity of the Architecture for the Action

Minimization of S_{Φ} is only possible for systems implementing the complete six-pillar architecture; the Entropic Action is not defined independently of it.

Proof. Minimizing S_{Φ} requires $\mathcal{D}$ strictly positive and finite, hence $\xi, \mathbf{K}, f > 0$ (equation (2)). By the pillar-to-factor dependencies established in section III B, this requires all six pillars simultaneously: Toll and Capacity sustain $\xi > 0$; Map sustains $\mathbf{K} > 0$; Protocol, Governor, and Margin jointly sustain $f > 0$. Removing any one

pillar collapses the corresponding factor to zero (reproducing the failure modes of section III C) so any trajectory with $\delta S_\Phi = 0$ that maintains structural coherence necessarily encodes the complete architecture. $\square$

The Entropic Action is therefore not defined independently of the architecture; the two are one closed claim. We now state the filter that determines how this minimization manifests in observation.

H. The Selection Principle

The Entropic Action defines *what* a system must minimize; we now state the filter that determines *how* minimization manifests in observation, reserving *which path* a surviving system follows for section IV. Configurations do not *seek* minimization; the principle is eliminative. What fails to minimize leaves no detectable signal, and what persists is the minimal-action remainder:

The **Selection Principle**: Among all configurations capable of encoding a given structural complexity, only those possessing the minimal possible **Entropic Action** (S_Φ) remain observable.

Because S_Φ counts elementary transitions, it is the cumulative ledger of a system's dissipation history. The substrate does not optimize this ledger; it erases ledgers that fail to balance.

IV. LEAST ACTION AS SURVIVORSHIP

The Entropic Action defines *what* a system must minimize; the Selection Principle establishes *that* only minimal-action configurations remain observable. We now derive *which* path a surviving system follows.

Standard quantum mechanics grounds the Principle of Least Action in phase interference. IE suggests a complementary, classical mechanism (thermodynamic elimination) with the substrate itself acting as a filter: what remains observable is only what has not yet dissipated.

A. The Exponential Filter

If persistence is maintained by state transitions, and each transition risks scrambling information into the substrate noise floor, then a history of N transitions survives only if every step survives. Independent errors compound multiplicatively, yielding an exponential filter.

Consider a structural history Γ composed of N elementary state transitions. Each transition scrambles a finite fraction of the system's distinguishable state information into the substrate's noise floor. Let q be the probability that a single transition maintains causal coherence rather than losing its informational trace to the background.

In Quiescent Steady State, successive transitions are statistically independent dissipative events. The probability that the trajectory Γ maintains coherence after N transitions is therefore strictly multiplicative:

$$P(\text{survival} | \Gamma) = q^N \quad (6)$$

The Entropic Action $S_\Phi[\Gamma]$ is the dimensionless count of elementary transitions accumulated over the history Γ . Rewriting q^N in exponential form, $q^N = e^{N \ln q}$, then substituting $N = S_\Phi[\Gamma]$ and $\beta \equiv -\ln q > 0$ (so β is positive, since $0 < q < 1$ makes $\ln q$ negative) gives $N \ln q = -\beta S_\Phi[\Gamma]$, turning equation (6) into:

$$P(\text{survival} | \Gamma) = e^{-\beta S_\Phi[\Gamma]} \quad (7)$$

Here $\beta \equiv -\ln q$ is the substrate's inverse entropic tolerance, a geometric invariant of the channel. For discrete substrates, q is determined by the channel geometry; in continuous media, β emerges from the substrate's noise spectral density.

The form itself is the informational analogue of the Boltzmann distribution $P \propto e^{-E/k_B T}$, with S_Φ playing the role of energy and β the role of inverse temperature. The Boltzmann form is derived from ensemble statistics at equilibrium and this form is derived from the multiplicative compounding of independent transition risk along a single history, presupposing neither equilibrium, a thermal bath, nor a physical substrate. Both converge on the same exponential law; the informational route recovers it under strictly weaker assumptions.

B. The Severity of the Selection Principle

How harsh is the Selection Principle? Let $\Gamma_{\min}$ be the minimal-action trajectory and Γ_{var} any variant path with a deviation $\delta S_\Phi > 0$. Over any observation window, the relative survival probability is:

$$\frac{P(\Gamma_{\text{var}})}{P(\Gamma_{\min})} = e^{-\beta \delta S_\Phi} \quad (8)$$

When the fundamental update cycle is much shorter than the observation timescale, even a small per-transition deviation compounds into a vast accumulated δS_Φ , and the survival ratio $P(\Gamma_{\text{var}})/P(\Gamma_{\min})$ falls to zero. The gap between the best trajectory and the next-best widens without bound.

C. No Merit Required

Severity is a statement about the losers, not the winner. It says nothing about whether $\Gamma_{\min}$ is efficient, elegant, or well-suited to anything, only that every alternative to it decayed faster. A mediocre trajectory survives

just as certainly as an excellent one, provided everything else was worse. What looks like design, foresight, or purpose is not a property of the winner; it is the absence of its competition.

D. The Principle of Least Action

The severity of the Selection Principle is formalized as:

Proposition 2 (Least Action as Survivorship). *In Quiescent Steady State, the observable trajectories of any persistent system are stationary paths of the Entropic Action:*

$$\delta S_\Phi = 0. \quad (9)$$

Proof. By the Selection Principle and equation (8), only trajectories minimizing S_Φ remain observable. The observable ensemble is therefore concentrated on the stationary manifold $\delta S_\Phi = 0$. $\square$

Standard classical mechanics takes the variational principle as foundational. Here, we demonstrate that this principle emerges naturally from a discrete informational substrate as the statistical boundary of survivorship. The record of what is left, not the rule being followed.

V. OPEN SYSTEMS: NECESSITY WITHOUT ADDITIVITY

Many persistent systems (cells, firms, ecosystems) are open, exchanging energy and information. But the necessity and sufficiency proofs of sections II and III were established strictly within Quiescent Steady State. We show necessity of the six pillars survives unchanged; only the Entropic Balance Sheet (section IIC) requires re-derivation in flux-driven capacity.

A. Pointwise Necessity and Operational Latency

Because the substrate continuously filters for causal coherence, survival is an instantaneous requirement, not a time-averaged one.

Proposition 3 (Pointwise Necessity). *For any system persisting past instant t , all six pillars must be structurally instantiated at t , regardless of external flux $d_e S(t)$.*

Proof. By the exponential filter (equation (7)), survival over a history Γ is strictly multiplicative. If any pillar is functionally absent during an arbitrarily small interval dt , the local survival probability drops to zero ($q_t \rightarrow 0$) via the failure modes of section IID. The entire multiplicative history collapses. The requirement is therefore instantaneous and flux-independent. $\square$

Flux alters a pillar's *operation*, not its *existence*. During phases of rapid growth or unbounded resource influx, a constraint mechanism might be operationally latent ($f_{\text{GOV}} \approx 0$), incurring no active coordination cost. Yet its *structural capacity* must exist at t ; if the mechanism is fundamentally absent, the system fails to persist the moment it reaches environmental saturation.

B. The Dynamic Budget: A Stochastic Master Equation for Shared Resources

Strict additivity ($Z_{\text{IE}} = \sum Z_i$) relies on Prigogine's near-equilibrium linear regime, where orthogonal domains prevent interference. In open systems, external flux raises the dissipation ceiling to a dynamic boundary $\mathcal{D}_{\text{max}}(t)$; as operations scale against this moving ceiling, resource contention forces non-linear cross-coupling ($\Delta(t)$) between previously orthogonal pillars:

$$Z_{\text{IE}}(t) = \sum_i Z_i(t) + \Delta(t) \quad (10)$$

We model contention between two of the six pillars, Protocol and Governor, as a stochastic service channel; the remaining four are not coupled in this treatment. Baseline transitions trigger the coordination demands each must process: Protocol routing, Governor constraint checks. By the Palm-Khinchine theorem, the superposition of many independent, sparse demands from distributed subsystem components converges to a Poisson arrival process, $\lambda \propto \mathcal{D}(t)$; the substrate serves these at capacity-limited rate $\mu \propto \mathcal{D}_{\text{max}}(t)$, exponentially distributed as the maximum-entropy prior for a known mean.

We define a shared First-Come, First-Served (FCFS) M/M/1 queue for these two streams; a minimal, analytically tractable model chosen to demonstrate cross-coupling in principle, not to capture the full statistics of real contention, and relaxed below via Kingman's heavy-traffic generalization to non-Markovian arrivals (GI/G/1).

The Shared-Stream Coupling: Protocol and Governor demands, with independent Poisson rates λ_{PRO} and λ_{GOV} , merge into a single FCFS M/M/1 stream $\lambda = \lambda_{\text{PRO}} + \lambda_{\text{GOV}}$, with utilizations $\rho_{\text{PRO}} = \lambda_{\text{PRO}}/\mu$, $\rho_{\text{GOV}} = \lambda_{\text{GOV}}/\mu$, and total $\rho = \rho_{\text{PRO}} + \rho_{\text{GOV}}$. This continuous-time Markov chain has exact stationary distribution $\pi_n = (1 - \rho)\rho^n$ and mean sojourn time $W = 1/(\mu(1 - \rho))$.

Applying Little's Law ($L_i = \lambda_i W$) to each demand stream decomposes the shared cost per pillar:

$$L_{\text{GOV}} = \lambda_{\text{GOV}} W = \frac{\rho_{\text{GOV}}}{1 - (\rho_{\text{PRO}} + \rho_{\text{GOV}})} \quad (11)$$

By Landauer's Principle, pending uncoordinated operations incur a continuous entropic cost ξ_0 ; the Governor's coordination cross-term is modeled as $\Delta_{\text{GOV}}(t) = \xi_0 L_{\text{GOV}}$.

This proves the cross-coupling within the shared-queue model: heavy Protocol routing ($\rho_{\text{PRO}} \rightarrow 1$) inflates the Governor’s entropic cost ($\Delta_{\text{GOV}} \propto 1/(1 - \rho_{\text{PRO}})$) even with static Governor demand. Within this model, the pillars cease to be orthogonal because they share a finite volumetric service capacity.

Macroscopic Limits and Generalizations: This model relates IE to three phenomenological regimes, two directly, one by expected extension:

1. **Onsager Recovery (Low ρ):** A van Kampen system-size expansion in the limit $\rho \ll 1$ is expected to recover linear force-flux relations ($\Delta \rightarrow 0$); QSS additivity is the leading-order behavior of the uncongested lattice.
2. **The Thrashing Asymptote (High ρ):** As $\mathcal{D} \rightarrow \mathcal{D}_{\text{max}}$, the rational divergence dominates: the system expends disproportionate action managing contention, forcing structural pauses (apoptosis, circuit breakers) to drop ρ and avoid causal collapse.
3. **Non-Markovian Variance (Kingman’s Formula):** Relaxing the memoryless assumption yields a general renewal system (GI/G/1), for which Kingman’s heavy-traffic approximation adds the coefficients of variation (c_a^2, c_s^2); our derived $\rho/(1-\rho)$ form is exactly its memoryless special case, explicitly isolating baseline geometric friction from domain-specific statistical variance.

The foundational claim of IE that the six pillars are structurally mandatory is entirely unaffected by the specific arrival distributions governing $\Delta(t)$, or by whether this two-pillar model extends to the full six. Within that scope, the Entropic Balance Sheet takes the form $Z_{\text{IE}} = \sum_i Z_i + \Delta(t)$, with the strict additivity of the Quiescent Steady State as the idealized limit where the substrate’s stochastic friction ($\Delta(t) \rightarrow 0$) vanishes.

VI. UNIVERSALITY MAPPING: ARCHITECTURE AND DYNAMICS

IE is identifiable across persistent systems of any substrate. By mapping both the static architecture (the six pillars) and the operational dynamics (the dissipation trade-off and selection filter), IE translates domain-specific phenomena into identical mechanical constraints. The following illustrative mappings prioritize conceptual clarity over technical completeness; while such operationalization belongs to applied IE and lies outside the scope of this baseline derivation, the underlying architectural invariant remains identical.

A. Biological Cell

Architecture: Capacity as metabolic hardware; Map as genome; Protocol as genetic code and signaling machinery; Governor as apoptosis and cell-cycle checkpoints; Toll as the thermodynamic cost of proofreading; Margin as ATP reserves and stress buffers.

Dynamics: The cell’s basal metabolic rate ($\mathcal{D}$) is the product of its ATP cost per molecular reaction (ξ), proteome complexity ($\mathbf{K}$), and metabolic turnover rate (f). An organism cannot simultaneously maximize genomic complexity and reproductive speed without exceeding the local thermodynamic ceiling. Trajectories that overdraw this budget undergo exponential suppression via starvation or competitive exclusion.

Cross-Domain Transfer: Rather than treating aging as a correlational pattern of stochastic damage, IE re-frames senescence as the degradation of operational pillars across nested sub-systems, giving biological decline the exact same structural failure signature as seen in other systems such as memory leaks and resource exhaustion in long-running software systems.

B. Ecological Network

Architecture: Capacity as biomass and solar energy capture; Map as trophic structure and niche topology; Protocol as nutrient cycling and metabolic exchange; Governor as carrying-capacity and predator-prey feedbacks; Toll as thermodynamic heat loss per trophic transfer; Margin as biodiversity and structural redundancy.

Dynamics: Total ecosystem respiration ($\mathcal{D}$) is the product of heat lost per transfer (ξ), food-web complexity ($\mathbf{K}$), and the rate of biomass turnover (f). Because solar influx rigidly caps $\mathcal{D}$, overly complex food webs with inefficient trophic transfers are systematically pruned by famine and ecological collapse, leaving only the least-action trophic topology.

Cross-Domain Transfer: Mapping ecological networks to energy grids and financial markets reveals that cascading collapse obeys the same fixed dissipation ceiling, making grid load-shedding and market circuit-breakers candidate tools for anticipating tipping points in biodiversity loss.

C. Large Language Model

Architecture: Capacity as inference compute and GPU memory bandwidth; Map as parametric weights and attention matrices; Protocol as tokenization, layer connections, and key-value (KV) caching; Governor as decoding hyperparameters (temperature, top- p), post-hoc alignment, and system prompt constraints; Toll as autoregressive latency and attention degradation per layer; Margin

as unused context window, attention head redundancy, and logit margin.

Dynamics: The model’s active dissipation burden ($\mathcal{D}$) is the product of compute friction per generated token (ξ), context and parameter footprint ($\mathbf{K}$), and generation velocity (f). In autoregressive generation, attempts to maximize context length and reasoning depth without lowering parameter activation costs hit a hard memory bandwidth and compute ceiling. The Entropic Action acts as a strict evolutionary filter on AI architectures: models that fail to minimize this product succumb to cumulative attention noise (hallucination) or catastrophic inference costs, driving the structural selection of minimal-action survivors (e.g., KV-cache distillation, sparse routing, or linear-time state-space models).

Cross-Domain Transfer: Giving the nascent field of AI access to cross-disciplinary knowledge without needing to re-derive it allows long-context reasoning degradation to be mapped directly to cellular metabolic exhaustion, suggesting cellular homeostatic regulation and neuro-vascular energy-capping protocols as candidate strategies for dynamic attention KV-cache management.

D. Further Applications

The same six-pillar mapping extends to domains further removed from physical substrates; two examples illustrate what the cross-domain transfer achieves in practice:

- **Institutional Organizations:** A shared dissipation metric shows how bureaucratic decay in a growing firm and signal decay in a fiber-optic network are directly comparable structural failures.
- **Scientific Enterprise:** Reframing Planck’s observation that science advances one funeral at a time as the thermodynamic survival of funded believers over verification friction shows that theoretical paradigms are constrained by the same dissipation limits as cells or firms.

VII. CONCLUSION

We have derived the minimal architecture of persistence and the mechanism that selects for it. The six pillars are necessary and sufficient; the Entropic Action is the filter; and the Principle of Least Action emerges as the accumulated boundary of what has not yet been forgotten.

Fluctuation theorems, master equations, and Langevin dynamics describe the statistics of transitions; IE describes the structure that must exist before any such

statistics can be defined, the discrete foundation that any continuous dynamics must presuppose. This shifts the fundamental question of complex systems from how existing structures optimize, to what minimal architectures are permitted to exist.

IE supplies something complex systems research has lacked: a formal boundary condition for persistence. Because IE maps every persistent system onto the same six-pillar substrate, it functions as a Rosetta stone across disciplines. Hard-won solutions in one domain become transferable theorems in another, the way protocol engineering that prevents signal decay in a fiber-optic network also prevents fragmentation in a corporate hierarchy. In emerging domains such as AI, where the pillars are present but unnamed, this vocabulary lets systems be engineered for resilience without re-deriving structural knowledge from scratch.

The same mapping lets persistent systems sharing an environment be ranked, not just described. A firm and a competitor, or two species in the same niche can be compared by how each manages its finite dissipation budget, the way two engines are compared by efficiency rather than by story.

Organizational decline, ecosystem collapse, and the limits of biological longevity are phenomena historically studied through domain-specific narrative and statistical correlation. By defining exactly what is required to persist, IE simultaneously establishes the mechanics of structural failure. Disparate phenomena are recast not as loose analogies, but as quantifiable structural failures under a single, unified mechanics.

Natural avenues for the extension include systems composed of nested persistent subsystems and a fuller account of decay itself: this paper treats termination as $\tau \rightarrow 0$, without characterizing the trajectory a system follows on the way there, or whether decay has its own structure worth deriving.

Informational Energetics does not explain why structure arises. It explains why only certain architectures remain to be observed. Persistence requires no intent, no optimization, and no forward-looking agency: only that the substrate be an unforgiving accountant, and that every ledger which fails to balance is erased. What remains is the architecture of persistence.

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